Min Chen

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EDUCATION

University of Pittsburgh — Pittsburgh, PA
Ph.D. in Computer Science | Advisor: Prof. Junyu Liu — *Sep 2025 – Present*

Georgia Institute of Technology — Atlanta, GA
M.S. in Electrical and Computer Engineering — *Sep 2022 – Jun 2025*

Tianjin University — Tianjin, China
B.E. in Functional Materials — *Sep 2018 – May 2022*

ONGOING PROJECTS

1. **DOE Genesis Mission — Agentic AI for Quantum Computing**
   Project Lead, Graduate Student Researcher — 2026 – Present

   - Investigating agentic AI for mathematical reasoning and theoretical discovery in measurement-based quantum computing (MBQC).

   `AI + Quantum` `AI + Math`

2. **Fujitsu Research Internship Project — AI Post-Training for Quantum Circuit Compilation**
   Research Intern — Patent application in progress; manuscript in preparation.

   - Developing diffusion models for quantum circuit synthesis; technical details are confidential while a patent application is pending, and a manuscript is in preparation.

   `AI + Quantum`

3. **FrontierPhysics — Benchmark for AI Agents in Frontier Physics Research**
   Core Contributor — Submitted to ICLR 2027; arXiv preprint in preparation.

   - Co-developing an open-source benchmark that evaluates how AI agents conduct end-to-end frontier physics research through iterative loops of literature review, research planning, implementation, and experimentation.
   - *Role:* Task author; contributed benchmark tasks drawn from my own research, took part in reviewing submitted tasks, co-developed the failure taxonomy and the failure-mode analysis of agent runs, and contributed to paper writing.

   `AI + Physics` `Agentic Benchmark`

4. **Tracepidia: Diagnosing Failures Behind Failures in Agent Trajectories**
   Co-first Author — Submitted to ICLR 2027; arXiv preprint in preparation.

   - Built a failure-diagnosis benchmark of 2,332 agent trajectories on 71 Terminal-Bench 2.1 tasks, each with one injected fault whose responsible component, failure category (41 in total), and error step are known.
   - Developed an auditor that combines the trajectory with runtime telemetry to attribute failures a trajectory-only judge cannot separate (macro-F1 0.315 to 0.424), and audited 890 agent sessions: 57.10% of verifier-passing sessions still contain a failure.

   `AI Safety + Agents`

5. **Grokking from a Scaling and Renormalization Group Perspective**
   First Author — Manuscript in preparation, targeting AISTATS 2027 and arXiv.

   - Developed a quantitative renormalization-group (RG) description of grokking: treating training epoch and weight decay as thermal and field coordinates, defining an order parameter, heat capacity, and susceptibility over an ensemble of modular-addition networks, constructing an effective real-space geometry of hidden neurons from a Hessian-derived coupling matrix, and performing block-spin RG whose linearized map yields critical exponents that match direct empirical estimates within 1–10%.

   `Interpretability from a Physics Perspective`

PUBLICATIONS

2026

1. **Min Chen**, Yu Gan, Xin Jin, Yuqing Li, Junqi Wang, Zeguan Wu, Yunfei Wang, Bingzhi Zhang, Priyam Srivastava, Tianlong Chen, Ankit Kulshrestha, Yuan Liu, Juan José Mendoza-Arenas, Kaushik P. Seshadreesan, Sarvagya Upadhyay, Xueyue Zhang, Quntao Zhuang, Junyu Liu.
When AI Meets Quantum Information: A Comprehensive Review.
arXiv preprint.
`AI + Quantum`

2. **Min Chen**, Jinglei Cheng, Pingzhi Li, Haoran Wang, Tianlong Chen, Junyu Liu.
Symbolic Analysis of Grover Search Algorithm via Chain-of-Thought Reasoning and Quantum-Native Tokenization.
npj Quantum Information, 2026.
`AI + Quantum`

3. Yakun Wang, **Min Chen**, Zeguan Wu, Junyu Liu, Sitao Zhang, Zhenwen Shao.
Infeasibility Aware Large Language Models for Combinatorial Optimization.
arXiv preprint, under review.
`LLM Bias + Optimization`

4. **Min Chen**, Zihan Wang, Canyu Chen, Zeguan Wu, Manling Li, Junyu Liu.
Artificial Entanglement in the Fine-Tuning of Large Language Models.
arXiv preprint, under Nature Computational Science submission.
`LLM Interpretability`

2025

1. **Min Chen**, Guansong Pang, Wenjun Wang, Cheng Yan.
Information Bottleneck-guided MLPs for Robust Spatial-temporal Forecasting.
International Conference on Machine Learning (ICML 2025).
`Interpretability for AI Post-Training`

2. David Meltzer, **Min Chen**, Junyu Liu.
Catapult Dynamics and Phase Transitions in Quadratic Nets.
Journal of Statistical Mechanics: Theory and Experiment, 2025.
`Interpretability from a Physics Perspective`

3. **Min Chen**, Bingzhi Zhang, Quntao Zhuang, Junyu Liu.
An Analytic Theory of Quantum Imaginary Time Evolution.
arXiv preprint, under Nature Computational Science submission.
`AI Interpretability for Quantum`

4. Haoran Wang, Pingzhi Li, **Min Chen**, Jinglei Cheng, Junyu Liu, Tianlong Chen.
GroverGPT: A Large Language Model for Quantum Searching.
arXiv preprint, under NPJ Quantum Information submission.
`AI + Quantum`

5. **Min Chen**, Minzhao Liu, Changhun Oh, Liang Jiang, Yuri Alexeev, Junyu Liu.
Towards Symmetry-Aware Efficient Simulation of Quantum Systems and Beyond.
IEEE TPS-ISA 2025.
`Perspective`

EXPERIENCE

Fujitsu Research of America — Santa Clara, CA
Research Intern — *May 2026 – Present*

- AI Post Training for Quantum Circuit Compilation.
- Co-authored a comprehensive review on the interplay of AI and quantum information (arXiv preprint, first author).

University of Pittsburgh — Pittsburgh, PA
Graduate Student Researcher — *Sep 2025 – Present*
Supervisor: Prof. Junyu Liu

- Investigated catapult dynamics and phase transitions in quadratic neural networks, establishing connections between statistical physics and deep learning (JSM 2025, second author).
- Derived an analytic theory for quantum imaginary time evolution, providing theoretical foundations for quantum learning algorithms (arXiv preprint, under submission, first author).
- Pioneered GroverGPT framework combining large language models with quantum algorithms, introducing quantum-native tokenization and chain-of-thought reasoning for quantum search (npj Quantum Information 2025, first author on GroverGPT-2; 1 arXiv preprint, under submission).
- Surveyed symmetry-aware simulation methods for efficient quantum system simulation, published as perspective paper at IEEE TPS-ISA 2025 (first author).
- Investigated LLM finetuning dynamics using tensor network tools from quantum information theory, providing explanations for observed phenomena through random matrix theory (arXiv preprint, under submission, first author).

University of Macau Macau, China
Research Assistant *Aug 2024 – Dec 2024*
Supervisor: Dr. Huanle Xu

- Designed and benchmarked LLM request scheduling strategies to optimize time per output token (TPOT), time to first token (TTFT), and throughput trade-offs.

AWARDS, GRANTS & FUNDED RESEARCH

- **U.S. Department of Energy Genesis Mission** 2026 – Present
- **Tinker Research Grant** 2025
  Thinking Machines Lab $5,000 USD
  Competitive research grant supporting independent research with compute credits and priority technical support.

TEACHING

- **CS 1503** Mathematical Foundations of Machine Learning *Teaching Assistant*
- **CMPINF 0401** Intermediate Programming *Teaching Assistant*

SERVICES

- **Reviewer**: KDD 2024; SLLM@ICLR 2025; *npj Quantum Information*; *Scientific Reports*; *ACM Transactions on Quantum Computing*; *Quantum Machine Intelligence*; *Quantum Information Processing*